

PRD: AI Note-Taking Assistant

1. Overview

A mobile-first AI agent designed to digitize, organize, and enrich handwritten notes and highlighted text from physical books, eliminating the need for manual transcription and organization.

2. Problem

- Users taking notes in physical books face high friction converting analog content into organized, searchable Markdown formats.
- Manual entry is time-consuming, and complex visual data like diagrams are often lost during transcription.
- The goal is to remove organizational overhead and ensure a "just snap a picture" experience for the user.

3. Goal

Create a high-signal capture experience that helps users archive tactical insights through natural language extraction and multimodal enrichment. This includes the seamless transformation of physical annotations into structured, digital archives.

4. Target User

- **Students and Academic Researchers:** Individuals who frequently interact with physical textbooks and journals.
- **Lifelong Learners:** Casual readers who prefer highlighting and marginalia but require a quick, organized digital archive for long-term retention.

5. Core Features (MVP)

5.1 Image Capture and Text Extraction

- **Mobile Interface:** A quick-capture interface allowing users to photograph physical book pages.
- **OCR & Element Extraction:** AI-powered identification and separation of highlighted book text and handwritten marginal notes.

5.2 Note Generation and Organization

- **Contextual Structuring:** Automatically inferring chapter or topic context from extracted text to organize notes without user intervention.
- **Markdown Formatting:** Generation of structured Markdown files that can be newly created or appended to existing archives.

5.3 Advanced Diagram Handling (Multimodal)

- **Diagram Segmentation:** Identification of illustrations, figures, or medical diagrams on the page.
- **Visual Recreation:** Using generative AI to recreate or enrich complex visuals with digital clarity and AI-generated explanations.
- **Reference Inclusion:** A fallback mechanism to maintain original images as references within the Markdown file if recreation is not desired.

6. Technical Requirements

- **Platform:** Web application, optimized for mobile browser responsiveness first, for high-quality image capture.
- **AI Models:**
 - **Vision/OCR:** Advanced multimodal models for detecting highlights and handwritten text.
 - **Orchestration:** Agentic frameworks to manage the "Think, Act, Observe" loop for note structuring.
 - **Image API:** External generative APIs for high-fidelity diagram recreation.

- **Storage:** Integration with cloud services like Google Drive or local secure storage for file persistence.

7. UI/UX

The design follows a minimalist approach to ensure the user focuses entirely on capture and high-quality digital output.

- **Quick-Capture Button:** A centered, prominent action button for immediate use.
- **Processing Feedback:** Clear visual states such as "Processing..." and "Notes Saved" to confirm successful capture.
- **Markdown Viewer:** A clean, readable in-app display of final digitized notes.
- **Visual Toggle:** A dedicated UI option to manage multimodal diagram handling preferences (recreate vs. reference).